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Visualizing Global Population Growth Trends

Data Visualization and Visual Analytics - Final Project

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Introduction

In this project, we aim to visualize a global insight into the population growth over time with significant shifts in population distribution, changing patterns of age structure, and rates of growth. By employing various data visualization techniques, our goal is to provide an elaborate view of how population dynamics changed and what were the main contributing factors behind these changes. The project evaluates several kinds of population data and explains, on the basis of regional distribution, age/age groups, and developmental levels, the key drivers behind population growth and their bearing on future survival.

- Analyze global population distribution: Using visualizations such as choropleth maps, explain where people are concentrated in the world.
- Examine historical trends: Research into how population changes operate by geography and by age groups over time.
- Understand growth rate dynamics: Use a line chart which relates the influence of birth and death rates on the growth of population.
- Compare development levels: Study only how are the growth rates of populations differing in a developed, as opposed to a developing region.
- Visualize future projections: Provide insights into expected future population trends and potential implications-concerning global resources and development.

User

The target users for this project including policymakers, demographers, researchers, and educators interested in global population trends. Targeting a large end-user potential, these visualizations would be made easy to understand for nonexperts and would also contain sufficiently detailed data to afford thorough analysis. These findings can improve policy outcomes in resource allocation decisions, urban planning strategies, and international development projects.

Website

[\[Data\]](#) [\[Charts\]](#) [\[Code References\]](#)

How many people live around the world today and where are they?

Figure 1: Population using Choropleth Map

Figure 2: Population by Region using Stacked Area Bar Chart

Figure 3: Population by Age Group using Stacked Area Bar Chart

Figure 4: Population by Births and Deaths Per Year using Line Chart

Figure 5: Population Growth Rate using Line Chart

Figure 6: Population Growth Rate by Level Development using Line Chart

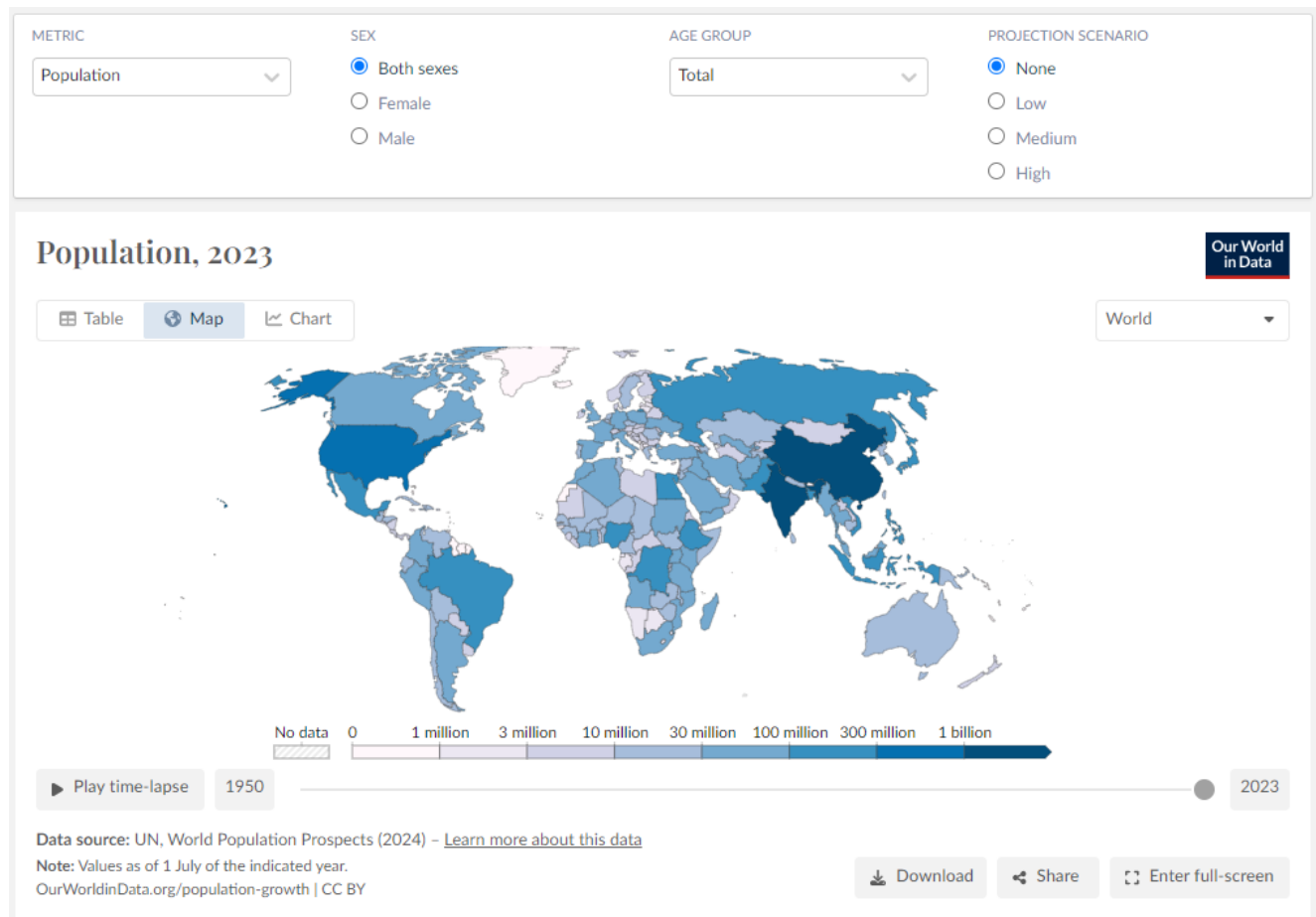


Figure 1: Population using Choropleth Map

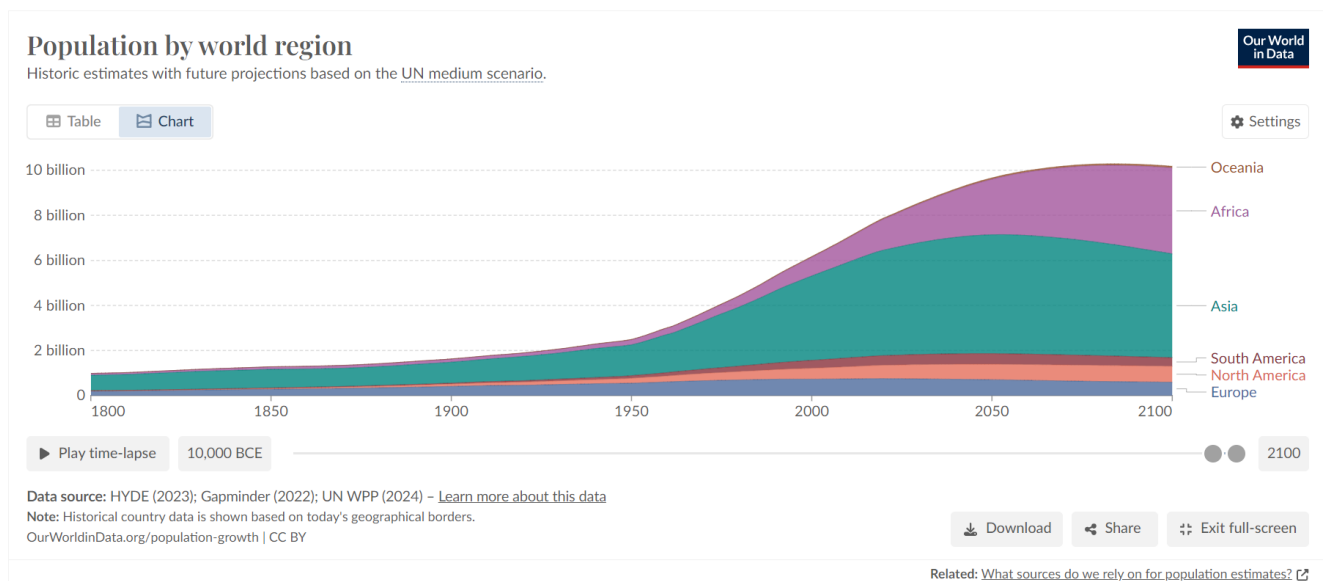


Figure 2: Population by Region using Stacked Area Bar Chart [\[Link\]](#)



Figure 3: Population by Age Group using Stacked Area Bar Chart

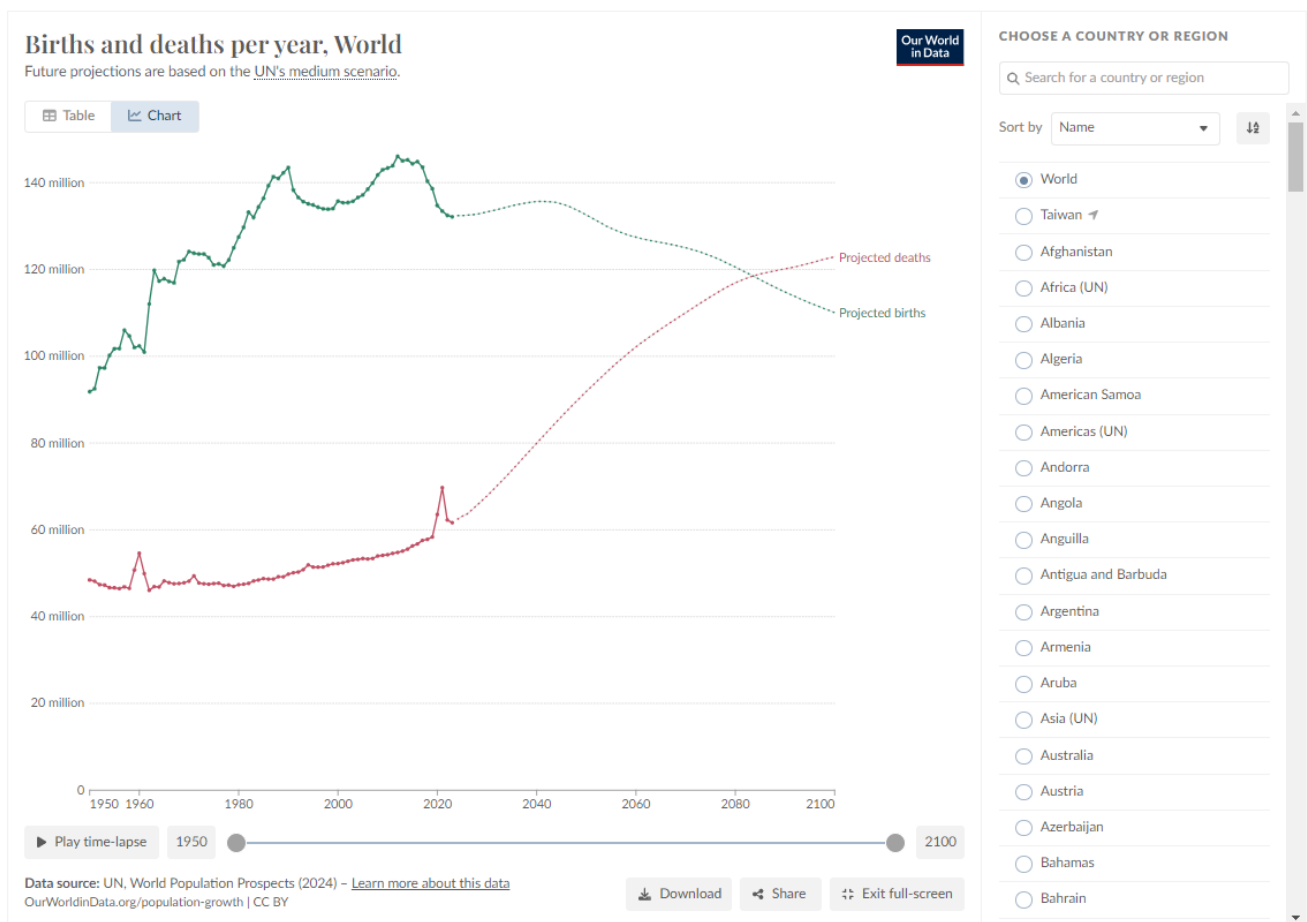


Figure 4: Population by Births and Deaths Per Year using Line Chart

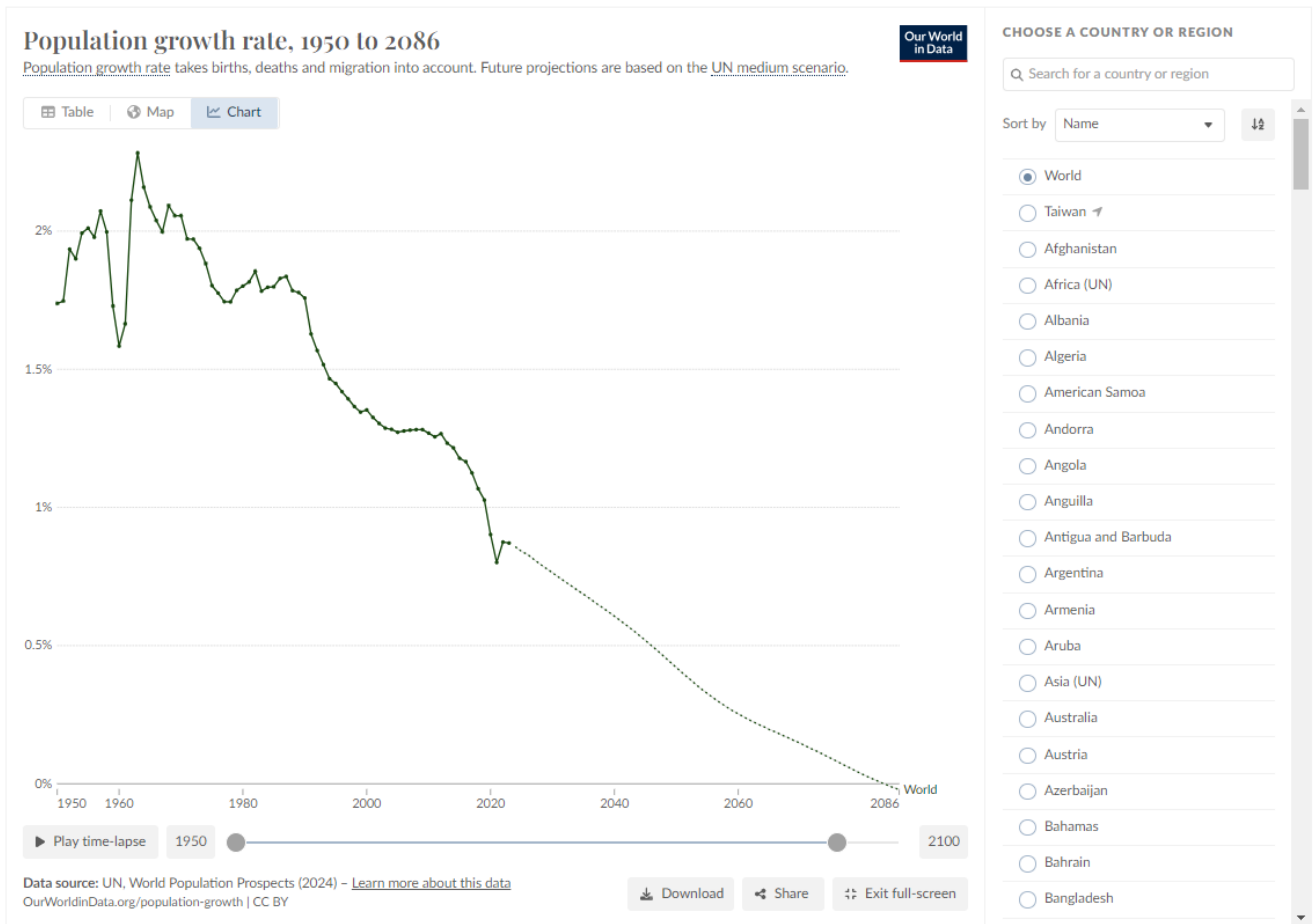


Figure 5: Population Growth Rate using Line Chart

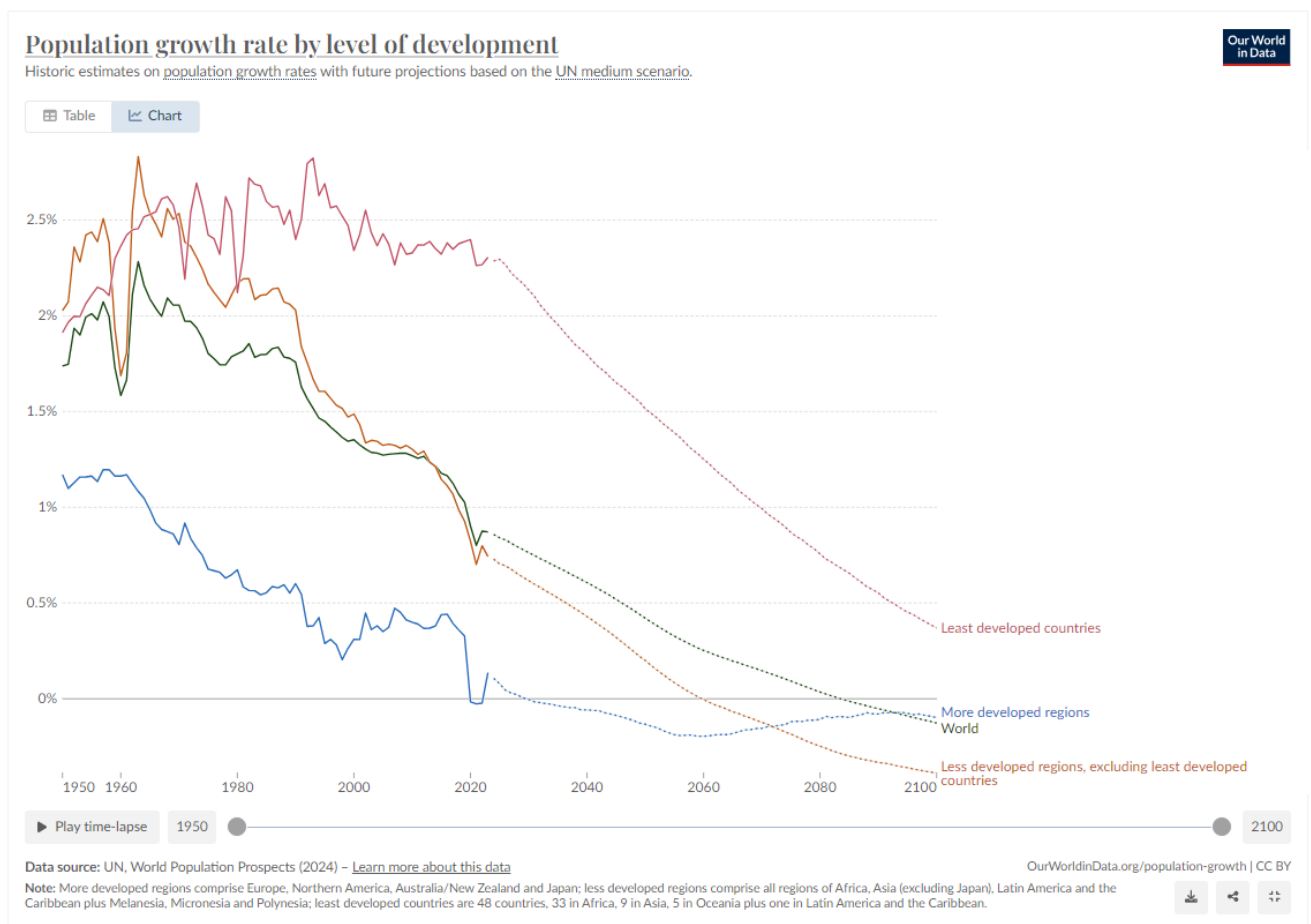


Figure 6: Population Growth Rate by Level Development using Line Chart